

Advanced AI for IT

Day 1 | Architect

OQ | Muscat | Sunday 27 September 2026

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TRACEZ

Training & Consultancy Services

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OQ | MUSCAT | 27 SEPT TO 1 OCT 2026

Advanced AI for IT

Five days of building AI systems that OQ's IT teams can own

5

days, 08:00 to 16:00

18

sessions, most of them
labs

2 to 3

people per capstone
group

1

working system per group
by Thursday

OQ | MUSCAT | 27 SEPT TO 1 OCT 2026

Pre-assessment:

<https://bit.ly/oq-pre-assess>

DAY 1 | WELCOME

Who is teaching this: production AI in industrial and regulated environments

INDUSTRIAL

MillMind

Production AI for paper mills: predictive maintenance, a knowledge base over plant documents, and an MCP layer connected to live distributed control systems

80,000+

professionals trained

REGULATED

Mashreq Bank, UAE

A production agentic AI system, from brief to deployment, in regulated financial services

20+

countries

PLATFORMS

AWS and Cerebras

Brought Amazon Bedrock and SageMaker JumpStart to market. Led generative AI strategy for Cerebras inference

20

production AI products shipped

DAY 1 | WELCOME

This week moves you from using AI tools to building AI systems

USING AI TOOLS

Prompt a chat window

Read the answer

Trust it or not

One person benefits



BUILDING AI SYSTEMS

Call a model from code

Parse a structured record

Measure it against a test set

A system others depend on

DAY 1 | WELCOME

Every AI system comes down to five decisions

1

Approach

Prompt, retrieve or tune?

2

Model

Which one, and what replaces it?

3

Hosting

API, managed cloud or self-hosted?

4

Integration

Which systems does it touch, and how?

5

Governance

Who approves it, watches it and stops it?

Your spec this afternoon has one section per decision.

DAY 1 | WELCOME

Five decisions, five days

Day	Decisions it settles	What you build
1 Architect	Approach, model, hosting	A spec and a cost model for one OQ system
2 Own the model	Model, hosting	A model you serve, stress and fine-tune yourself
3 Ground it	Approach	Retrieval over manuals, procedures and diagrams
4 Agents	Integration	An agent with tools, an approval gate and a trace
5 Integrate and ship	Integration, governance	An MCP server, a deployment pack, your capstone

DAY 1 | WELCOME

What you leave with: a working capstone and eight artifacts

Your capstone

A working AI integration for one OQ IT use case, with source code, architecture, a deployment plan and a governance checklist

- | | | | |
|---|------------------------------------|---|---------------------------------------|
| 1 | AI Architecture Decision Framework | 2 | RAG System Blueprint, text and images |
| 3 | Agent Design Patterns Library | 4 | Custom MCP Server Package |
| 5 | Production AI Deployment Checklist | 6 | Capstone System Package |
| 7 | Tuned model and evaluation | 8 | Document Vision Pack |

DAY 1 | WELCOME

How the week runs

1

Labs run in Colab

One link per lab. Nothing to install on your laptop

2

Groups when possible

Mixed skills, one system per group

3

A saved result for every lab

If something breaks, we show it and keep moving

4

Breaks built into the day

Coffee and lunch

DAY 1 | AGENDA

By 16:00 you each have a spec and a cost model for one OQ system

08:00		Welcome and pre-assessment	Assessment	30 min
08:30		Introductions and program arc	Talk	20 min
08:50		Get connected: Colab and the API key	Setup	15 min
09:05	S1	Fundamentals	Lab	70 min
10:15		Break		15 min
10:30	S1	How models are trained and run	Talk	20 min
10:50	S2	The September 2026 model landscape	Talk	35 min
11:25	S3	Build, buy or host	Talk	35 min
12:00		Dhuhr and lunch		45 min
12:45	S4	Architecture spec and cost model	Lab	90 min
14:15	S5	Spec peer review	Discussion	25 min
14:40		Capstone groups form	Facilitated	20 min
15:00		Day 2 prep	Setup	15 min
15:15		Break and Asr		20 min
15:35		Claude Code demo, then close	Demo	25 min

DAY 1 | S1 FUNDAMENTALS

Five questions and one terminal check decide how we spend the next 80 minutes

HANDS UP IF YOU HAVE

- 1 Called a model API from your own code?
- 2 Parsed JSON that a model returned?
- 3 Used embeddings or a vector store?
- 4 Built anything with tool calling?
- 5 Run a model on your own machine?

Mostly yes

Faster through the basics, more time on Part C

Mostly no

Every slide, at a slower pace

DAY 1 | S1 FUNDAMENTALS

Run the environment check: green means go, red means we fix it now

```
python setup/setup_check.py
```

DRY RUN

- ✓ Python version
- ✓ Package install works
- ✓ API key present, test call succeeds
- ✓ Local model server reachable
- ✓ Drive mounts
- ✓ Disk space
- ✓ GPU present or absent

<https://github.com/AIGuruOne/OOQ-labs>

DAY 1 | S1 FUNDAMENTALS

Tokens are the unit of cost, latency and context

1

Cost

You pay per million tokens, in and out, at different prices

2

Latency

Output is generated one token at a time, so long answers are slow

3

Context

The window a model can see is measured in tokens, not pages

Rule of thumb for English: about 4 characters, or three quarters of a word, per token. Arabic usually costs more tokens per word. Measure yours.

Try it: [OpenAI Tokenizer: platform.openai.com/tokenizer](https://platform.openai.com/tokenizer) | [tiktoken: github.com/openai/tiktoken](https://github.com/openai/tiktoken)

DAY 1 | S1 FUNDAMENTALS

A context window is a budget, not storage

WHAT A 200-PAGE HSE MANUAL COSTS TO SEND



The fix

Send the three pages that answer the question, not all 200. That is retrieval, Day 3.

At 1,000 questions a day: about 133 million input tokens a day, before a single word of the answer.
That is about \$27 a day on the cheapest efficient model, and \$1,330 on the top frontier tier.

DAY 1 | S1 FUNDAMENTALS

Temperature is a product decision: triage wants zero, drafting wants some

Triage

Temperature 0

Same ticket in, same record out. Consistency is the product

OQ: Routing INC-004412 to a queue

Drafting

Temperature above 0

Some variety reads more naturally and gives reviewers options

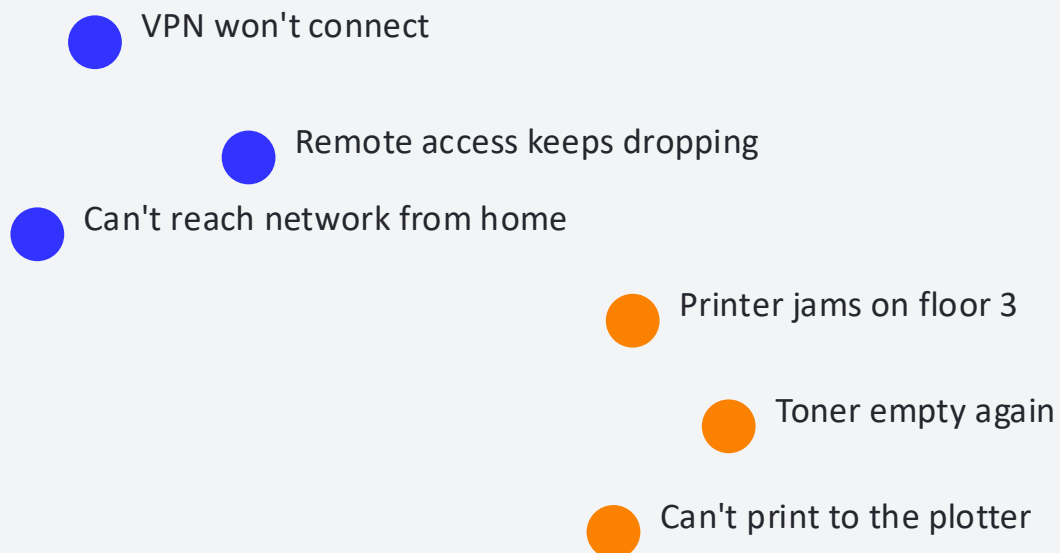
OQ: Wording a shift handover summary

Zero reduces variation. It does not guarantee identical output, so test for it.

Try it: [Transformer Explainer](https://poloclub.github.io/transformer-explainer), temperature slider: poloclub.github.io/transformer-explainer

DAY 1 | S1 FUNDAMENTALS

Embeddings turn meaning into distance



Schematic: real embeddings have hundreds of dimensions

How it works

A model turns text into a list of numbers. Similar meanings get similar numbers, so "close" means "about the same thing".

Try it: [Embedding Projector: projector.tensorflow.org](https://projector.tensorflow.org) | [IBM: ibm.com/think/topics/vector-embedding](https://ibm.com/think/topics/vector-embedding)

DAY 1 | S1 FUNDAMENTALS

Similar tickets land close together even when they share no words

"VPN keeps dropping when I work from home"

"Remote access disconnects every few minutes"

Shared words: none that matter. Keyword search misses this pair.

DRY RUN

Live notebook output

Similarity scores for this pair and an unrelated ticket, from 01_fundamentals

Read: [OpenAI embeddings guide: developers.openai.com](https://openai.com/blog/openai-embeddings)

DAY 1 | S1 FUNDAMENTALS

Structured output turns a model into a component other systems can call

FREE TEXT

"This looks like a network issue, probably VPN. I'd say it's fairly urgent since month end is close."

A person can read it. Code cannot rely on it.

SCHEMA-CONSTRAINED

```
{ "category": "network",  
  "urgency": "high",  
  "impact": "single_user" }
```

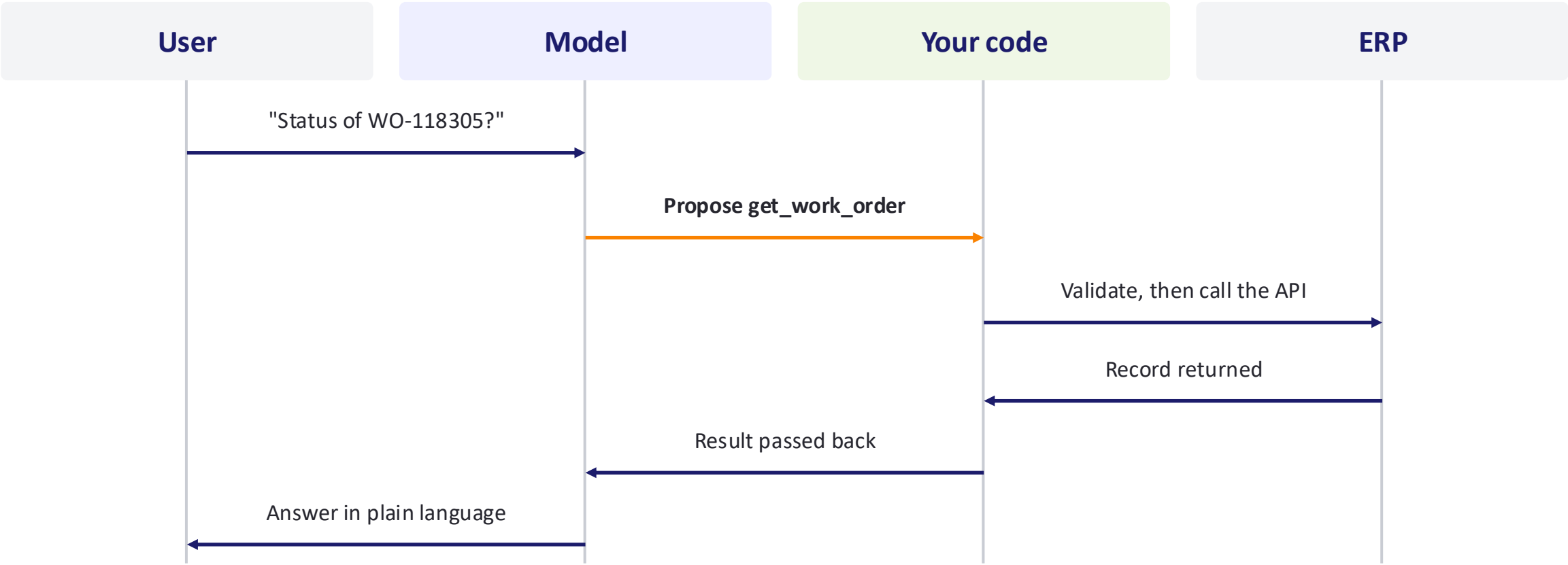
Fixed fields, allowed values, always parses.



Service desk system
creates and routes the ticket

Read: [OpenAI structured outputs: developers.openai.com](https://openai.com/blog/openai-structured-outputs)

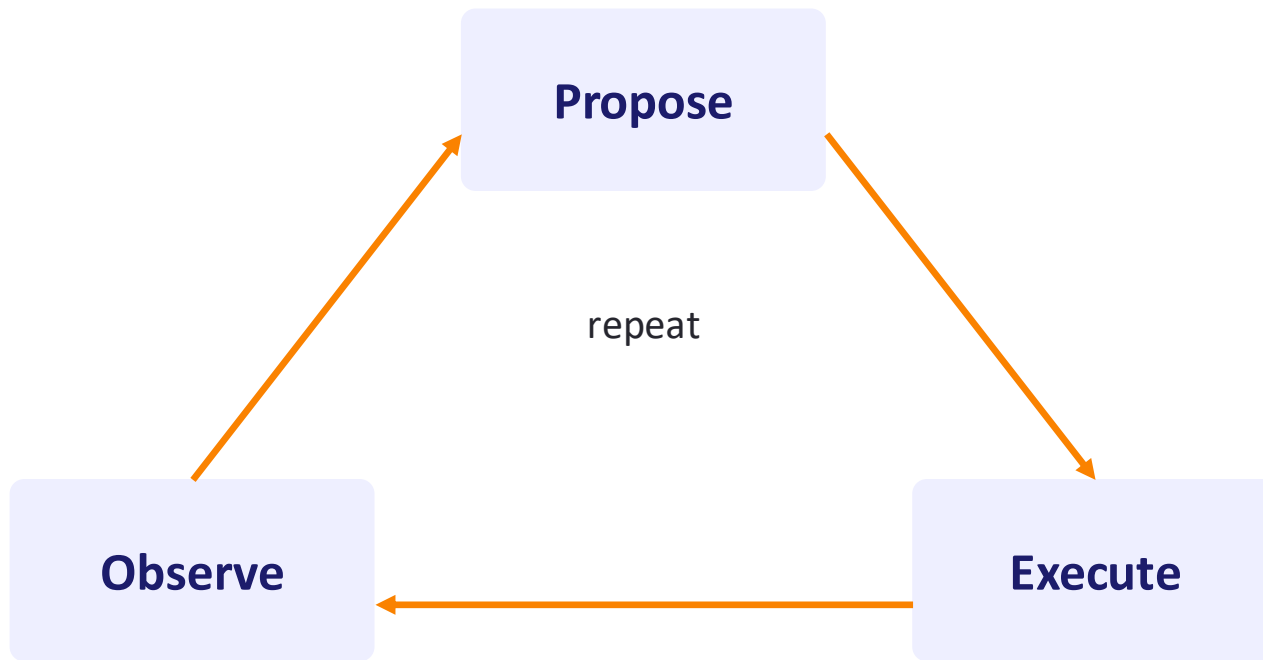
Tool calling: the model proposes, your code executes



Read: [Claude tool use: platform.claude.com/docs](https://platform.claude.com/docs) | [OpenAI function calling: developers.openai.com](https://developers.openai.com)

DAY 1 | S1 FUNDAMENTALS

An agent is a loop: propose, execute, observe, repeat until a stop condition



Stop when

- ✓ The goal is met
- ✓ The step budget runs out
- ✓ The cost budget runs out
- ✓ A human says stop

Read: [Anthropic, Building effective agents: anthropic.com/engineering](https://anthropic.com/engineering)

DAY 1 | S1 FUNDAMENTALS

The whole agent fits on one screen, and each part maps to a day this week

```
messages = [SYSTEM_PROMPT, request]
for step in range(MAX_STEPS):
    reply = model.call(messages, tools=TOOLS)
    if reply.tool_call is None:
        return reply.text
    if reply.tool_call.name in WRITE_TOOLS:
        require_approval(reply.tool_call)
    result = run_tool(reply.tool_call)
    log(step, reply.tool_call, result)
    messages += [reply, result]
raise StepBudgetExceeded
```

`model.call` Day 2: your own model

`messages` Day 3: retrieved context

`TOOLS, run_tool` Days 4 and 5: MCP

`require_approval` Day 4: the approval gate

`log, MAX_STEPS` Day 5: governance

DAY 1 | S1 FUNDAMENTALS

System prompts are configuration: version them like code

PROMPTS/TRIAGE_SYSTEM.MD V1.3 TO V1.4

- Set urgency to high if the user says urgent.
- + Set urgency from business impact, not from
- + the user's wording. "Urgent" alone is medium.

One sentence changed. Every urgency score on the service desk changes with it.

Treat it like code

- ✓ Kept in git
- ✓ Changed by review
- ✓ Tested on the eval set first
- ✓ Rolled back like a release

Prompt, retrieve, tune: three tools for three different problems

Tool	Fixes	When it wins	What it costs	OQ example
Prompt	Unclear instructions	Always the first thing to try	Minutes; a few tokens per call	Tighter triage rules in the system prompt
Retrieve	Missing knowledge	Facts that change or must be cited	An index to build and keep fresh	Answering from the current HSE procedure
Tune	Inconsistent format or behavior	A fixed pattern at high volume	Data, training, retraining per model change	Ticket to a seven-field record

Read: [OpenAI, Optimizing LLM accuracy: developers.openai.com](https://openai.com/blog/optimizing-llm-accuracy)

DAY 1 | S1 FUNDAMENTALS | WHAT GOES WRONG

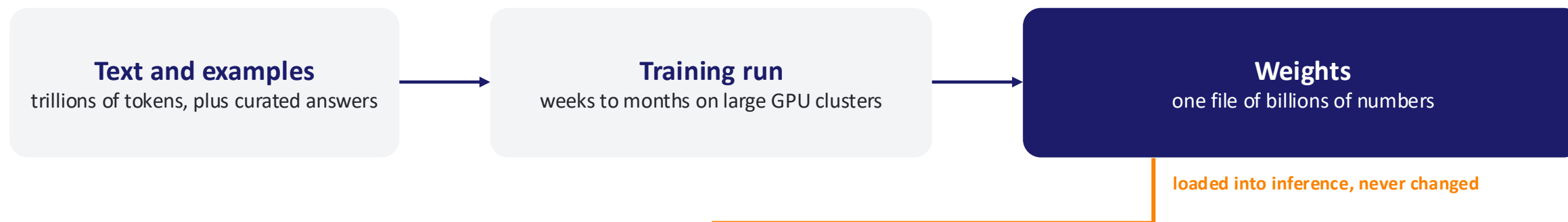
Missing fields, truncated context and runaway loops: you will meet all three this week

Failure	What causes it	What catches it	Where you meet it
A required field comes back empty	Free text where a schema was needed	Schema enforcement and validation	Day 2
The answer ignores the key paragraph	Context truncated past the window	Retrieve less, measure what was sent	Day 3
An agent calls the same tool forever	No step or cost budget	MAX_STEPS, budgets, a kill switch	Day 4

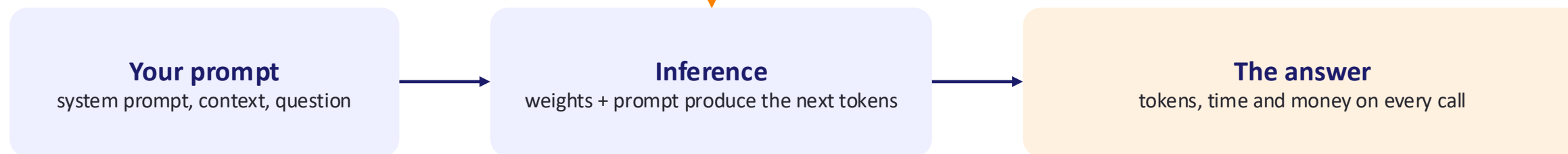
DAY 1 | S1 FUNDAMENTALS | HOW MODELS ARE TRAINED AND RUN

Training makes the weights once; inference uses them on every call

TRAINING | ONCE, BY THE VENDOR



INFERENCE | EVERY CALL, BY WHOEVER HOSTS IT



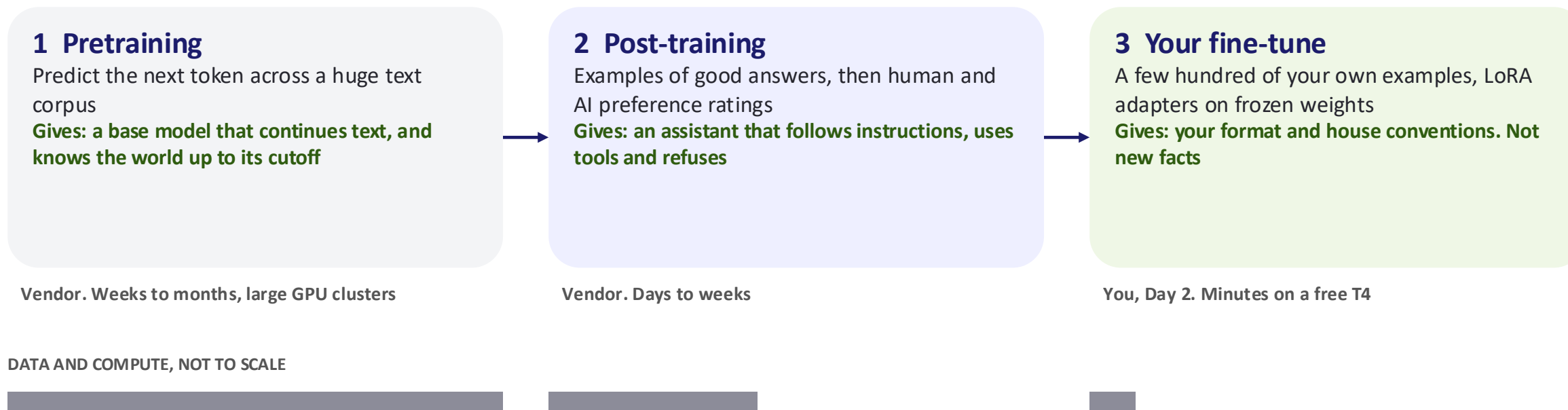
loaded into inference, never changed

Everything you buy this week is inference: tokens, GPUs, latency. The only training you will do is a small fine-tune on Day 2.

Watch: [3Blue1Brown, LLMs explained briefly: youtube.com](https://www.youtube.com/watch?v=3Blue1Brown)

DAY 1 | S1 FUNDAMENTALS | HOW MODELS ARE TRAINED AND RUN

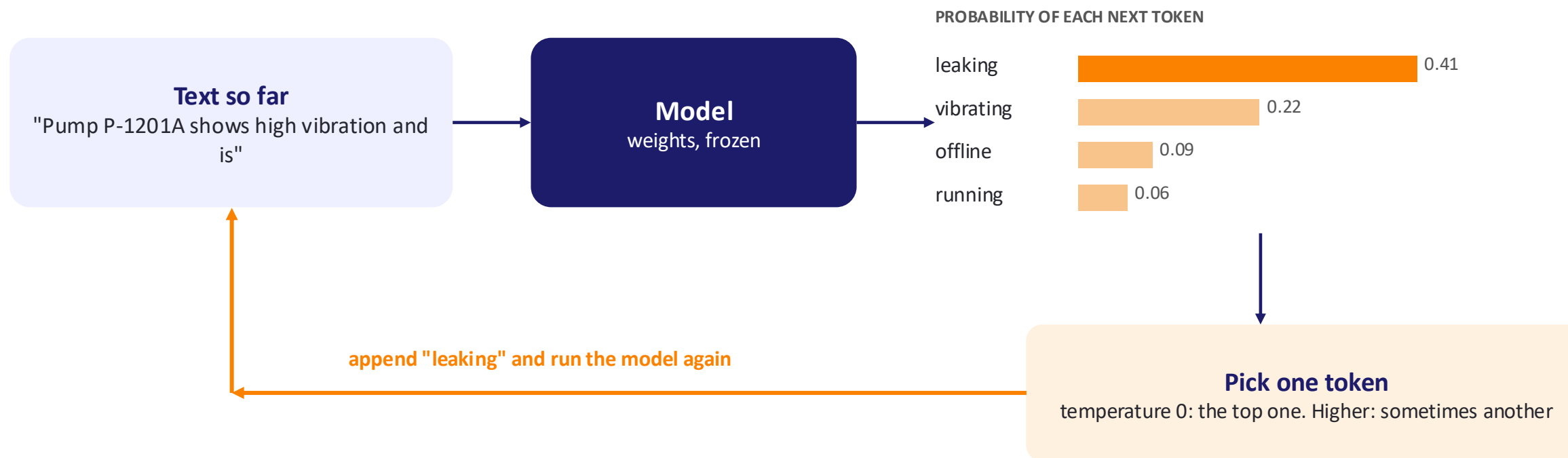
Training has two stages: pretraining learns language, post-training learns to be an assistant



Knowledge cutoff: the date the pretraining data stops. After it, the model knows only what you put in the prompt. That is retrieval, Day 3.

Watch: [Karpathy, Intro to LLMs: youtube.com](https://www.youtube.com/watch?v=KjZKw333333) | [LoRA: huggingface.co/docs/peft](https://huggingface.co/docs/peft)

Inference is a loop: predict one token, add it to the text, repeat



One model call per output token. A 300-token answer is 300 passes through the weights. Probabilities are illustrative.

Try it: [Transformer Explainer: poloclub.github.io/transformer-explainer](https://poloclub.github.io/transformer-explainer)

DAY 1 | S1 FUNDAMENTALS | HOW MODELS ARE TRAINED AND RUN

Reading your prompt is fast; writing the answer is slow

PREFILL: THE WHOLE PROMPT IN ONE PASS



1 step, all tokens in parallel

DECODE: THE ANSWER, ONE TOKEN PER STEP



8 tokens = 8 steps, each waiting for the last

Output costs more

Output tokens are priced 5 to 6 times input on the S2 price table. Short answers are cheap answers

Latency has two numbers

Time to first token (prefill) and tokens per second (decode). Day 2 measures both

Ask for less text

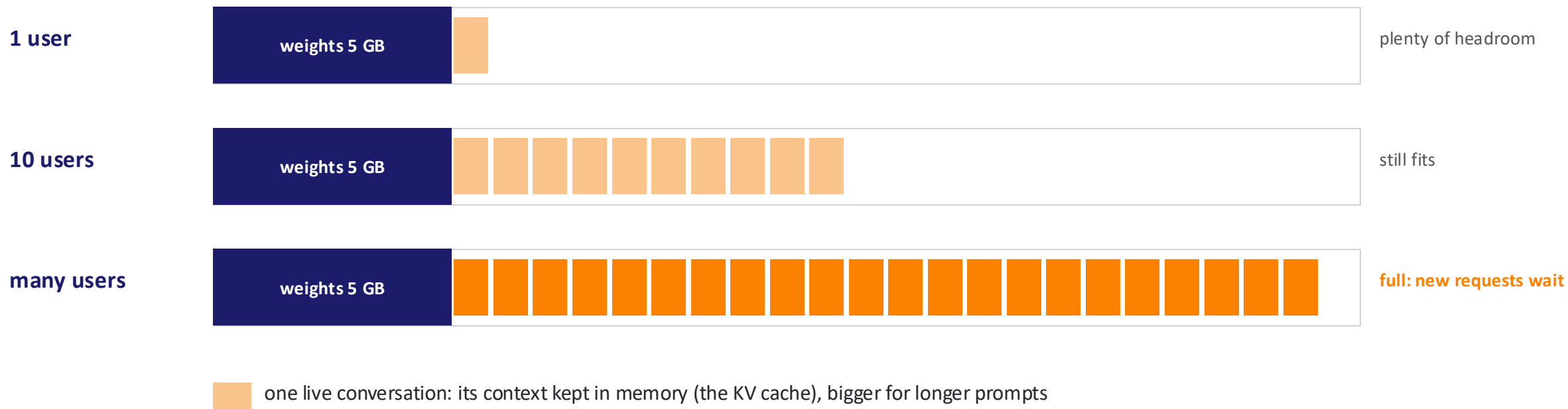
A JSON record of 60 tokens beats a paragraph of 400: faster, cheaper, easier to check

Read: [NVIDIA, LLM inference optimization: developer.nvidia.com/blog](https://developer.nvidia.com/blog/llm-inference-optimization/)

DAY 1 | S1 FUNDAMENTALS | HOW MODELS ARE TRAINED AND RUN

GPU memory holds the weights plus every conversation in flight

ONE 24 GB GPU SERVING AN 8B MODEL AT 4-BIT (ILLUSTRATIVE)



Memory, not speed, usually caps how many people one server can serve. Longer prompts fill it faster. Day 2 S8 finds the limit on a real GPU.

Read: [Hugging Face, KV cache: huggingface.co/blog/kv-cache](https://huggingface.co/blog/kv-cache) | [vLLM PagedAttention: vllm.ai/blog](https://vllm.ai/blog)

DAY 1 | S1 FUNDAMENTALS | HOW MODELS ARE TRAINED AND RUN

What OQ IT owns: inference always, training almost never

Activity	Who does it	Does OQ IT do it?	Where this week
Pretraining	The model vendor or lab	Never	Talked about, today
Post-training	The model vendor	Never	Talked about, today
Fine-tuning with LoRA	You, when a fixed format at volume justifies it	Sometimes	Day 2 lab
Serving the model (inference)	The vendor API, or a VM you run	Always decides; sometimes runs	Day 2, Day 4
Evaluating the answers	You, on your own data	Always	Days 2 to 5

The skill this week builds is choosing, serving and checking models, not making them.

DAY 1 | S2 THE SEPTEMBER 2026 MODEL LANDSCAPE

Since June: new frontier models, new prices and new access gates

When	What changed	Why it matters for OQ
9 June	Anthropic releases Claude Fable 5 and Mythos 5	A new tier priced above the old frontier: \$10 in, \$50 out per million tokens
12 June to 1 July	Both suspended to comply with US export controls, then restored	Access can change for reasons outside the vendor and outside OQ
Since June	Claude Opus 5.5 at \$4 / \$20, below Opus 5 at \$5 / \$25	Frontier capability gets cheaper inside the same tier
Until 31 December	Gemini 3.8 Flash on a promotional price	Budget at the price after the promotion: \$1.50 / \$7.50
22 September	OpenAI GPT-6 Astra: limited rollout, off by default for enterprises	The newest model may not be available to you yet

Sources: [Anthropic statement: anthropic.com/news/fable-mythos-access](https://anthropic.com/news/fable-mythos-access) | [OpenAI: openai.com/index/gpt-6-astra](https://openai.com/index/gpt-6-astra)

Three tiers matter for OQ: frontier, efficient and open-weight

1

Frontier

Hardest reasoning, long documents, agents. \$2 to \$10 per million input tokens

Use it where quality pays

2

Efficient

Triage, extraction, summaries: most enterprise volume. \$0.20 to \$1 per million input tokens

Start here, prove you need more

3

Open-weight

Runs on a machine you control. You pay for GPUs and people, not tokens

When data cannot leave

DAY 1 | S2 THE SEPTEMBER 2026 MODEL LANDSCAPE | DECISION TABLE

The same 1,000 queries a day cost \$30 or \$1,350 a month, depending on the model

Tier	Model	Per million tokens, in / out	Per month at 1,000 queries a day
Frontier	OpenAI GPT-6 Astra (limited rollout)	\$10 / \$50	\$1,350
Frontier	Claude Fable 5.1	\$10 / \$50	\$1,350
Frontier	Claude Opus 5.5	\$4 / \$20	\$540
Frontier	Gemini 3.1 Pro	\$2 / \$12	\$300
Efficient	Claude Haiku 4.5	\$1 / \$5	\$135
Efficient	OpenAI gpt-5.4-mini	\$0.75 / \$4.50	\$113
Efficient	Gemini 3.8 Flash (promotional price)	\$0.75 / \$3.75	\$101
Efficient	OpenAI gpt-5.6-luna	\$0.20 / \$1.20	\$30

Compare: [Artificial Analysis: artificialanalysis.ai](https://artificialanalysis.ai) | [Claude pricing: platform.claude.com/docs](https://platform.claude.com/docs)

DAY 1 | S2 THE SEPTEMBER 2026 MODEL LANDSCAPE

Some frontier models are gated, and one was switched off for nearly three weeks

Gated at launch

Claude Mythos Preview goes only to Project Glasswing partners. GPT-6 Astra started with a limited set of organizations, off by default for enterprises.

Switched off by regulation

Fable 5 and Mythos 5 were suspended from 12 June to 1 July to comply with US export controls. Access returned when the controls were lifted.

What to plan

Build on a model you can buy today. Keep a tested second route, another vendor or an open-weight model, behind the same interface.

DAY 1 | S2 THE SEPTEMBER 2026 MODEL LANDSCAPE

Open-weight models are sized by memory: about 0.6 GB per billion parameters at 4-bit

1

1 to 3 billion

1.3 to 2 GB. A laptop or a free Colab T4 runs it. Tomorrow's lab

2

7 to 8 billion

About 5 GB. One 24 GB GPU, with room for many users' context

3

70 billion

About 43 GB. Two 24 GB GPUs or one 80 GB GPU

Then add the context cache: every concurrent conversation needs memory too. Families to test: Llama, Qwen, Gemma, Mistral. Check each license before production.

Browse: [Ollama library, with download sizes: ollama.com/library](https://ollama.com/library)

Capability differs most on long documents, strict schemas and diagrams

Workload	What differs	What goes wrong	How to test it	OQ example
Long documents	Usable context, not advertised context	Details in the middle get missed	Plant a fact on page 150, ask for it	A 200-page HSE manual
Strict schemas	Native structured output support	Missing or invented fields	50 real records, count schema failures	Ticket to a seven-field record
Diagrams	Vision quality on dense drawings	Tag numbers not on the drawing	Score extracted tags against a key	Tags from a P&ID

DAY 1 | S2 THE SEPTEMBER 2026 MODEL LANDSCAPE

Arabic and bilingual documents: test on your own content, not a benchmark

Benchmarks

Tell you where to start

Public test sets, mostly English and general chat. Your document types are rarely in them

OQ: A leaderboard rank for Arabic chat

Your own test

Tells you what to buy

20 real documents in Arabic, English and both. Score against a key you wrote

OQ: Bilingual HSE procedures and ticket text

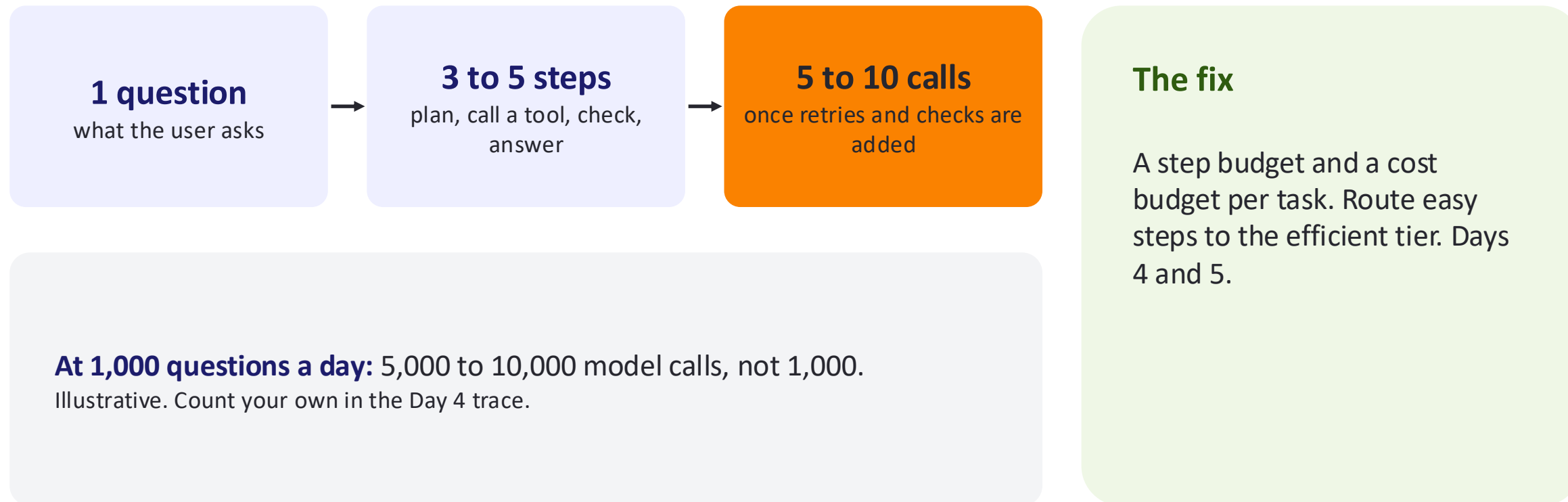
Arabic usually costs more tokens per word, so the price comparison changes too. Measure both.

Compare: [Open Arabic LLM Leaderboard: huggingface.co/spaces/OALL](https://huggingface.co/spaces/OALL)

DAY 1 | S2 THE SEPTEMBER 2026 MODEL LANDSCAPE

Price per token keeps falling, but agents and retries multiply the calls

ONE USER QUESTION, AS AN AGENT RUNS IT



Read: [Anthropic, Building effective agents: anthropic.com/engineering](https://anthropic.com/engineering)

DAY 1 | S2 THE SEPTEMBER 2026 MODEL LANDSCAPE

Models retire: plan the replacement on the day you choose one

.ENV THIS WEEK'S LABS, PINNED ON PURPOSE

```
HOSTED_MODEL=gpt-4o-mini  
# Day 3 and 4 notebooks: gpt-4.1-mini  
# Frozen 24 Sept. Same model, known outputs
```

Older than today's price table, on purpose: a lab that changes under you teaches nothing.

Plan it on day one

- ✓ Name the replacement model
- ✓ Keep the eval set that proves it
- ✓ Budget a retest per retirement
- ✓ Pin the version in config

Check: [Claude deprecations: platform.claude.com/docs](https://platform.claude.com/docs) | [OpenAI deprecations: developers.openai.com](https://developers.openai.com)

DAY 1 | S2 THE SEPTEMBER 2026 MODEL LANDSCAPE

Five questions to answer before you pick a model

ANSWER THESE BEFORE ANY BENCHMARK

- 1 What is the most sensitive data it will ever see?
- 2 How many requests a day, and at what peak?
- 3 Free text, or a fixed schema?
- 4 Which languages and inputs: Arabic, scans, diagrams?
- 5 Which model replaces it, and who retests?

Then

Test two or three candidates on 20 of your own cases

Decide

The cheapest model that passes your bar

Compare: [Arena leaderboard: arena.ai](https://arena.leaderboard.ai) | [Artificial Analysis: artificialanalysis.ai](https://artificialanalysis.ai)

DAY 1 | S3 BUILD, BUY OR HOST

Three ways to put a model into production

1

Vendor API

The provider runs the model.
Your data travels to them

Fastest start

2

Managed cloud

The model runs inside your
cloud tenancy and region

Your contract, your region

3

Self-hosted

The model runs on a machine
you control

Your hardware, your team

DAY 1 | S3 BUILD, BUY OR HOST | DECISION TABLE

Each hosting option wins somewhere, and each has a bill

Option	When it wins	What it costs	OQ example
Vendor API	Public or internal data, a pilot, frontier capability	Per-token pricing; data leaves your boundary	Drafting from public vendor manuals
Managed cloud	Confidential data inside an approved tenancy	Per-token pricing plus cloud commitments; regional model gaps	Service desk triage in OQ's cloud region
Self-hosted	Restricted data, high steady volume, version control	GPUs, people, patching and on-call	Extraction from restricted engineering drawings

DAY 1 | S3 BUILD, BUY OR HOST

Data classification drives this decision more than capability does

Public Published manuals, vendor datasheets

Internal Service desk tickets, IT runbooks

Confidential Maintenance history, HSE incident reports

Restricted Engineering drawings of live assets

Ask first:
what is the most
sensitive data this
system will ever
see?

Example scheme. Use QQ's own classification in your spec.

Each data class narrows the hosting options that can carry it

Data class	Vendor API	Managed cloud	Self-hosted
Public	✓	✓	✓
Internal	Check terms	✓	✓
Confidential	X	If the tenancy is approved	✓
Restricted	X	X	✓

Example mapping

OQ security owns the real one. Ask for it before you build, not after.

DAY 1 | S3 BUILD, BUY OR HOST

Residency is a per-model, per-region question: check before you commit

- 1 Is this exact model offered in our region?
- 2 Where is inference actually processed?
- 3 Where are prompts and logs stored, and for how long?
- 4 Is any of our data used for training?

Why per model

Providers roll out new models to some regions first. Your region can lag the announcement by months.

Read: [Microsoft Foundry deployment types \(Global, Data Zone\): learn.microsoft.com](https://learn.microsoft.com/en-us/fabric/fabric-foundation-model-deployment-types)

DAY 1 | S3 BUILD, BUY OR HOST

At 1,000 queries a day, the API bill is small and predictable

ASSUMPTIONS, STATED

Queries a day	1,000
Input tokens per query	2,000
Output tokens per query	500
Days a month	30

Per month

60 million input tokens

15 million output tokens

27 SEPT

Efficient tier: \$30 to \$135 a month. Frontier: \$300 to \$1,350

60 × input price + 15 × output price, per million tokens, as of 27 September

DAY 1 | S3 BUILD, BUY OR HOST

At 100,000 queries a day, self-hosting can beat frontier prices, not efficient ones

API, per month

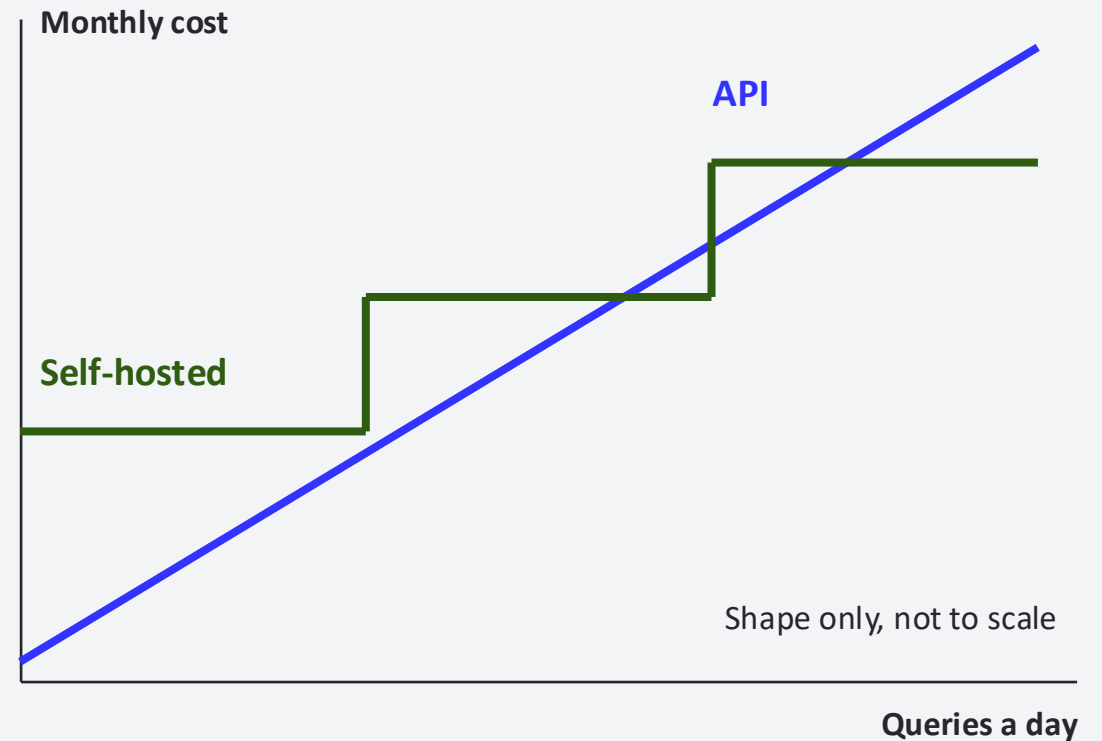
\$3,000 to \$135,000, efficient to frontier

Grows in a straight line with volume

Self-hosted, per month

GPU servers needed × monthly cost, plus people

Grows in steps, one server at a time



DAY 1 | S3 BUILD, BUY OR HOST

Caching and routing move the cost more than the choice of model does

1 Prompt caching

A long prefix that repeats, such as a system prompt or a manual, is billed at a reduced rate after the first call

OQ: The same 20-page policy in front of every question

2 Routing

A cheap check sends easy requests to a small model and only hard ones to a frontier model

OQ: Password resets to the small model, multi-system incidents to the large one

Cache reads bill at 2.5 to 10% of the input price at Anthropic, 10% at OpenAI. Batch jobs are half price at both.

Read: [Claude prompt caching: platform.claude.com/docs](https://platform.claude.com/docs) | [OpenAI prompt caching: developers.openai.com](https://developers.openai.com)

DAY 1 | S3 BUILD, BUY OR HOST

The hidden cost of self-hosting is people, patching and on-call

What gets budgeted: GPU servers or cloud GPU hours

WHAT ALSO NEEDS AN OWNER

Model and runtime updates

Security patching

Capacity planning for peaks

Monitoring and alerts

An on-call rota

Re-evaluation on every update

DAY 1 | S3 BUILD, BUY OR HOST | WHAT GOES WRONG

A pilot built on an API that security will never approve for production

What happened

A team builds a ticket assistant on a vendor API with real tickets. Users love the pilot. Security review blocks production: tickets contain confidential data.

What caused it

Nobody checked the data class at design time. The hosting choice was made for speed, and the pilot had no path to production.

What catches it

A data class field in every spec, filled before the first line of code. The hosting option is chosen from it, not before it.

DAY 1 | S3 BUILD, BUY OR HOST

Your turn: fill the hosting matrix for your own system

Your system	Most sensitive data	Data class	Hosting option	The one reason

12 minutes, on your own. Open OQ_D1_S3_Decision_Matrix.xlsx: gates first, then score the four options for one system. We hear each choice before lunch.

Two truths and a hallucination

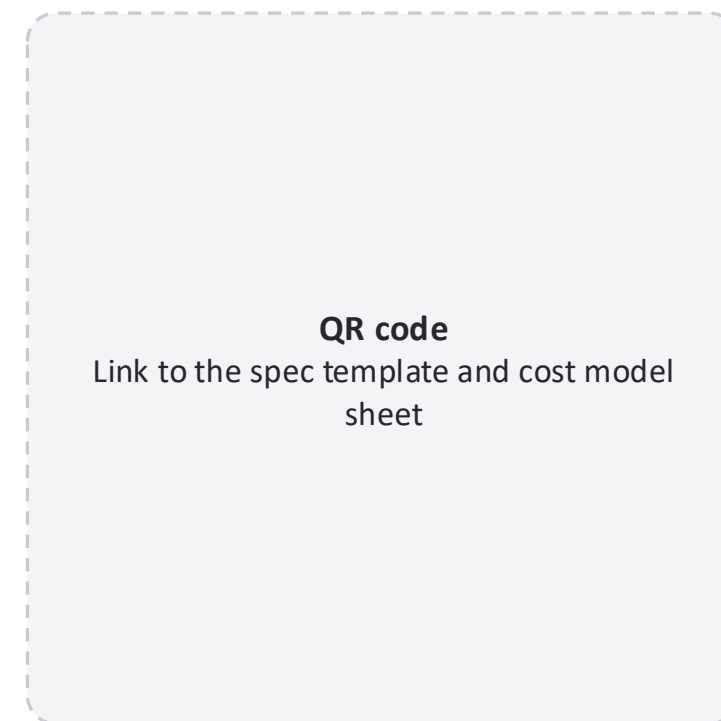
- 1 Everyone writes three statements about AI at OQ: two true, one invented but plausible
- 2 Read them to your table. The table votes on which one is invented
- 3 One table reads its best invented statement to the room

Why this one: A plausible invention is harder to catch than an obvious error. That is the week's problem in one game.

DAY 1 | S4 ARCHITECTURE SPEC AND COST MODEL | LAB

Draft a spec and a cost model for one OQ system

TEMPLATES	Spec template, cost model sheet
TIME BOX	90 minutes
RUNS ON	Your laptop. No notebook, no GPU
WORK IN	Pairs, one system per pair



Scan to open the templates

You are building one spec with five decisions, and a cost model beside it

1 Approach Prompt, retrieve or tune?

2 Model Which one, and what replaces it?

3 Hosting API, managed cloud or self-hosted?

4 Integration Which systems does it touch, and how?

5 Governance Who approves it, watches it and stops it?



Cost model

Tokens per query, in and out
Monthly cost at 1,000 a day
Monthly cost at 100,000 a day
Your recommendation, in one line

Five steps, in the order the template asks for them

1

Name the system

One sentence: who uses it, for what

2

Classify the data

The most sensitive data it will ever see

3

Make the five decisions

One reason each, not a paragraph

4

Fill the cost model

Tokens per query, then both volumes

5

Write the recommendation

What you would build first, and why

Checkpoint: five decisions made, and a cost at both volumes

At 13:30

- ✓ Data class filled
- ✓ All five decisions have one reason
- ✓ Hosting follows from the data class

At 14:05

- ✓ Tokens per query estimated
- ✓ Cost at 1,000 and 100,000 a day
- ✓ Recommendation in one line

If you are stuck, start from the worked example

The worked example

A completed spec and cost model for service desk triage, in the same templates

The one-line fallback

Copy the example, change the system and the data class, and make the five decisions again

DAY 1 | S4 ARCHITECTURE SPEC AND COST MODEL | DEBRIEF

The hardest decision is the one your spec will be challenged on

Which of the five decisions was hardest to make, and why?

This feeds S5 at 14:30. Another pair reviews your spec against the same five decisions.

DAY 1 | S5 SPEC PEER REVIEW

Review the spec you are given against the five decisions, not against your own

Decision	Ask the authors
Approach	Why this approach and not the other two?
Model	What replaces it when it is retired?
Hosting	Does it follow from the data class?
Integration	Which systems does it write to, and who approves the writes?
Governance	Who can stop it, and how fast?

10 minutes each way. Write one question per decision on the spec.

The assistant that passed the demo and failed the review

WHAT HAPPENED

An IT team builds a ticket assistant on a vendor API in three weeks. Users love the pilot. Real tickets were used throughout, and some contained incident detail. Security blocks production: that API is not approved for confidential data. The team is asked to "just move it somewhere else".

IN YOUR GROUP

1

Which of the five decisions was made too late?

2

What would you have asked in week one?

3

Can the pilot be saved, or is it a rebuild? Say why.

DAY 1 | CLOSE

Choose one of five OQ use cases for your capstone

- | | | |
|---|--|---|
| 1 | Service desk triage | Ticket in, routed record out |
| 2 | Technical documentation assistant | Cited answers from manuals and procedures |
| 3 | HSE incident intelligence | Patterns across incident reports |
| 4 | ERP work order agent | Reads and drafts work orders, with approval |
| 5 | Operational reporting | Daily summaries from structured data |

DAY 1 | CLOSE

Groups of two or three, with complementary skills, building one system

1

Two or three people

Small enough that everyone codes

2

Complementary skills

Mix integration, data and application people

3

One system

The use case you choose now, all the way to Thursday

Five minutes: stand by the use case poster you want, then balance the groups.

DAY 1 | CLOSE

By Thursday your group will have a working system, not a slide about one

Day 2

A model you can serve and tune

Day 3

Retrieval over your documents

Day 4

An agent with an approval gate

Day 5

An MCP server, a deployment pack, a live demo

Thursday, 14:00: 12 minutes per group. Problem, architecture, live demo, deployment plan.

DAY 1 | CLOSE

Tomorrow: own the model. Arrive with Colab open and signed in

WHAT WE BUILD

A model served on a machine you control

A load test that shows where it breaks

A fine-tuned model for ticket triage

HAVE READY AT 08:30

✓ Colab open and signed in

✓ Your spec, with the hosting decision

✓ Your capstone group, seated together